

Light-Sensitive FIRE ALARM

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This light-sensitive fire alarm circuit is extremely sensitive, low-cost and easy to assemble even by a beginner. It uses infrared photo transistor as fire sensor.

Circuit and working

Circuit diagram of the light-sensitive fire alarm is shown in Fig. 1. It is built around L14F1 photo transistor (T1), BC547 transistor (T2) and NE555 timer (IC1).

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- T1 - L14F1 photo transistor
- T2 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 150-kilo-ohm
- R2 - 4.7-kilo-ohm
- R3 - 1-kilo-ohm
- R4 - 56-kilo-ohm

Capacitors:

- C1 - 0.047 μ F ceramic disk
- C2 - 0.01 μ F ceramic disk
- C3 - 47 μ F, 25V electrolytic
- C4 - 100 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- S1 - On/off switch
- LS1 - 8-ohm, 0.5W speaker
- 9V battery

T1 can sense both infrared and visible light. NE555 based astable multivibrator is used for audible warning. Resistors R3 and R4 and capacitor C1 decide the frequency. Here, frequency is around 272Hz, which can be changed by changing values of R3, R4 or C1.

When T1 senses light, T2 stops conducting. As a result, reset pin 4 of IC1 is held high. This makes the multivibrator to function and an alarm to sound in loudspeaker LS1.

Construction and testing

An actual-size, single-side PCB of the light-sensitive fire alarm is

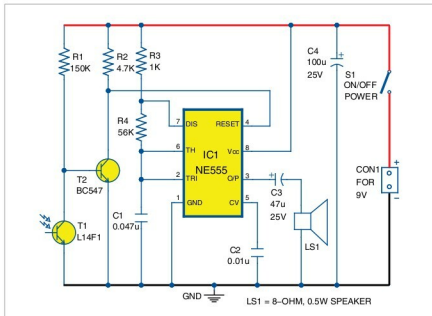


Fig. 1: Circuit diagram of the light-sensitive fire alarm

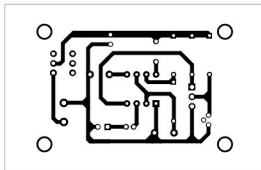


Fig. 2: Actual-size PCB pattern of the light-sensitive fire alarm

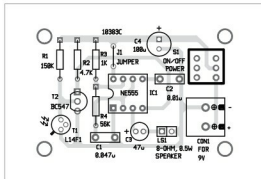


Fig. 3: Component layout of the PCB

shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable plastic case.

Make a small hole in the plastic case for the photo transistor (L14F1) so that fire flame is detected/sensed easily through it. T1 sensor should not face directly towards normal light sources like sunlight, bulbs or tubelights to avoid a false alarm.

The circuit works off a 9V battery. **EFY**

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